The Cellular Data Center:

An In-Silico, Falsifiable Blueprint for Language Machines Built on Living Neural Tissue

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Abstract

Living neurons cultured on multi-electrode arrays can now be stimulated, recorded and rented remotely, and in 2022 a culture of this kind showed apparent learning of a simplified Pong game within five minutes of closed-loop play. These results invite an ambitious question: could living neural tissue serve as a computational substrate for language modelling? We answer it with a proposal and a test rather than a claim. We introduce the *Cellular Data Center*, a modular architecture in which a silicon front end encodes tokens as stimulation patterns, racks of living cultures (input, a proposed recurrent "memory ring", a proposed association culture, and output) supply rich nonlinear dynamics, a bioreactor schedules perfusion and rest, a silicon back end decodes spikes into the next token, and a homeostatic critic delivers predictable or unpredictable feedback. Its first version is a reservoir hybrid: only a linear readout is trained. We state five falsifiable hypotheses, anchored to one strict null model: *a random echo state network of equal size, with the same readout and the same budget, does at least as well*. We give a simulation-first protocol that any group can run before touching tissue. Finally, we report a small, fully reproducible pilot ($N{=}500$ units, 5 seeds). In it, the null model is not rejected. A frozen leaky integrate-and-fire reservoir retained 0.80 units of linear memory capacity against 49.2 for the equal-size echo state network. Slower synapses raised this to 1.44, at the cost of next-character accuracy, and a naively wired ring did not help. We treat this negative result as the paper's most useful finding: it pinpoints time scale, not neuron count, as the first engineering problem for wetware language machines.

1 Introduction

Large language models are built on transformers [1] and run on GPU and TPU data centers. Their energy footprint is now a policy-level concern. The International Energy Agency estimates that data

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centers used about 415 TWh of electricity in 2024, around 1.5% of global consumption, and projects that this will more than double to about 945 TWh by 2030 [2]. Training GPT-3 alone was estimated at 1,287 MWh [3]. The human brain, by contrast, runs on roughly 17–20 W [4]. At that rate, one megawatt would power fifty thousand brains.

That contrast has pulled biology back into computing. In *DishBrain*, cortical neurons on a high-density multi-electrode array were embedded in a simulated Pong game and showed apparent learning within five minutes of real-time play. No such learning appeared in control conditions [5]. *Brainoware* used a brain organoid as an adaptive reservoir for speech recognition and nonlinear-equation prediction [6]. FinalSpark opened a remotely accessible platform that has hosted more than one thousand organoids [7]. In March 2025 Cortical Labs released the CL1, a commercial unit that keeps neurons alive on a silicon chip with built-in life support; remote cloud access is also offered [8]. The research program now has a name, *organoid intelligence* [9].

The language question remains open, and it is easy to answer badly. Three temptations stand out. The first is scale: a living network of "billions of neurons" is not on the table, because DishBrain-style cultures are seeded with about 10^6 cells per array [10]. The second is mechanism: it is tempting to assume that a culture "learns language" when the silicon decoder may be doing the work. A widely shared 2026 demonstration of cultured neurons playing *Doom* illustrates the risk. Its own code notes that the device "performs no computation" and that the game still plays with a randomized decoder [11]. The third is training by force, with high-voltage "punishment" for wrong answers. The published closed-loop paradigm instead uses unpredictable stimulation at amplitudes in the same safe range as the reward signal [5, 10].

This paper takes a different route. We propose an architecture, and we make it easy to prove wrong.

Contributions.

1. **Architecture.** The *Cellular Data Center* (Section 3, Figure 1) is a modular, scale-out design. Silicon handles the parts silicon is good at (tokenization, encoding, decoding, scheduling), and living tissue supplies high-dimensional, fading-memory dynamics. Every module is labelled by evidential status (Table 1).
2. **A strict null model and five falsifiable hypotheses** (Section 4). The central test is simple. If the biology-style substrate cannot beat a random echo state network of the same size under an identical readout, it is not contributing.
3. **A simulation-first protocol** (Section 5) with fixed tasks, baselines, ablations and reporting rules, executable on a laptop with open tools.
4. **A reproducible pilot** (Section 6) that runs the null-model test at small scale and reports a negative result in full, with code.

2 Background

Reservoir computing. A reservoir is a fixed, high-dimensional dynamical system driven by input. Only a linear readout is trained [12–14]. Its usefulness rests on two properties: nonlinear expansion of the input, and *fading memory*, so that the present state still carries a decaying trace of the recent past. Jaeger's linear memory capacity, $\mathrm{MC} = \sum_{k\geq 1} r^2(u_{t-k}, \hat{u}_{t-k})$, measures the second property [15]; it is bounded by the number of state variables [16]. Because training touches only the readout,

almost any physical system with these properties can serve as a reservoir [17]. That is what makes the paradigm attractive for tissue, which cannot be back-propagated through.

Cultured networks as reservoirs. Dissociated cortical networks on electrode arrays already behave like reservoirs. Dranias et al. found that stimulus-specific information remained decodable for about one second after stimulation and that network bursts erased these hidden memories [18]. Brainoware made the analogy explicit with organoids [6]. Both use the logic we adopt: the tissue provides dynamics, and silicon learns to read them.

Closed-loop learning and the free-energy principle. The free-energy principle holds that self-organizing systems act to reduce surprise about their inputs [19]. Cultured cortical neurons have been shown to perform blind source separation in line with this account [20]. DishBrain turned the idea into a training signal. A successful interception was followed by predictable stimulation (75 mV, 100 Hz, 100 ms across the eight sensory electrodes). A miss was followed by unpredictable stimulation (150 mV, 5 Hz, at random sites, for 4 s, then a 4 s rest) [10]. A later preprint reports that, in time-matched comparisons, these cultures were more sample-efficient than DQN, A2C and PPO agents [10]. We adopt this two-signal scheme for our critic and do not use stronger stimulation.

Spiking and neuromorphic language models. Silicon spiking systems are the natural control group. Neuromorphic processors such as Loihi run spiking networks with on-chip learning [21]. SpikeGPT showed that a generative language model can be trained with spiking units [22]. Our proposal differs in kind: the recurrent core is not trained at all, and in the target system it would be alive.

Platforms. FinalSpark's Neuroplatform runs four multi-electrode arrays, each holding four organoids with eight electrodes per organoid. Organoid lifetimes reach about 100 days in the best cases, and access through a Python API is free for research [7]. The CL1 retails for about US$35,000 and keeps neurons alive for up to six months [8]. Trade press reports about 800,000 neurons per unit and 59 stimulation inputs [23]; we treat these figures as unverified vendor-level descriptions.

3 The Cellular Data Center

3.1 Components

Front end (silicon). A tokenizer maps text to symbols $x_t \in \{1, \ldots, V\}$. An encoder E maps each symbol to a spatiotemporal stimulation pattern $\mathbf{s}_t = E(x_t) \in \mathbb{R}^{M \times \tau}$ over M electrodes and a presentation window of τ milliseconds. Version 1 uses a small vocabulary ($V \leq 32$) and fixed random electrode footprints per symbol, the wetware analogue of an embedding table.

Core (tissue). The culture's state $\mathbf{r}_t$ evolves as

$$\mathbf{r}_t = \Phi(\mathbf{r}_{t-1},\, \mathbf{s}_t;\; \theta_{\text{bio}}(t))\,, \tag{1}$$

where θ_{bio} collects synaptic and cellular parameters that the experimenter cannot set directly and that drift with plasticity, fatigue and age. In a rack, four cultures are chained by microfluidic channels: input, a ring-shaped recurrent culture intended as working memory, an association culture, and output. Racks $2, \ldots, N$ repeat the layout and run in parallel. The data-center analogy is literal here: capacity grows by adding racks, not by growing a single culture.

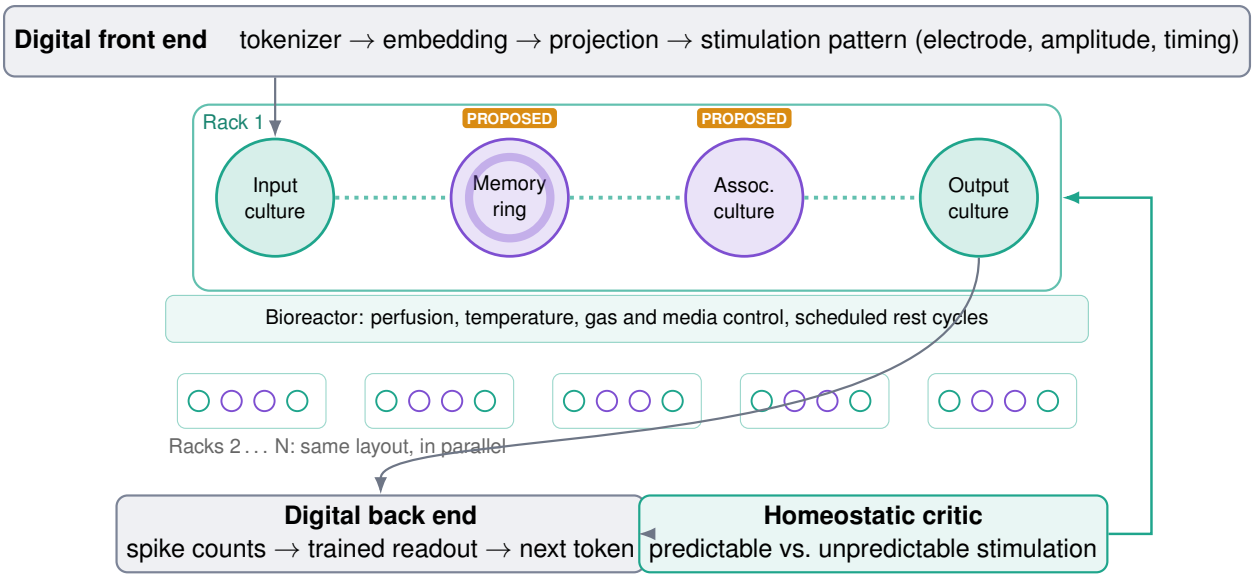


Figure 1: **The Cellular Data Center.** Grey: silicon. Teal and purple: living tissue (simulated in this paper). Tokens are encoded as stimulation patterns, cultures supply dynamics, a trained readout decodes the next token, and a critic returns predictable stimulation after correct outputs and unpredictable stimulation after errors (right-hand return path). The memory ring and association culture are hypotheses, marked PROPOSED. Version 1 uses a single culture and trains only the readout.

Back end (silicon). Spike counts in time bins form a feature vector $\phi(\mathbf{r}_t)$. A readout W produces next-token scores, $p(x_{t+1} \mid x_{\leq t}) = \mathrm{softmax}(W\phi(\mathbf{r}_t))$. Only W is trained by gradient descent or ridge regression.

Critic and life support. After each prediction a critic delivers either a structured, predictable stimulus (correct) or an unpredictable one (incorrect), following the DishBrain scheme and its amplitudes [5, 10]. A bioreactor under each rack handles perfusion, temperature, CO_2 and media, and imposes rest periods. These rest periods mean the system cannot serve tokens continuously, and an honest energy account must include them.

3.2 What this proposal does not claim

We record four boundaries so that the design cannot be quoted beyond its evidence. (i) No culture of "billions of neurons": planning assumes 10^5–10^6 cells per culture, or 10^3–10^4 simulated neurons. (ii) No high-voltage punishment: feedback amplitudes stay in the published range. (iii) The memory ring, any oscillation-based token code, and the idea that potentiation alone yields semantics are hypotheses, not mechanisms. (iv) No claim that cultured neurons have played *Doom* in a sense that rules out the decoder doing the work [11].

4 Hypotheses and the null model

Let $\mathcal{S}$ be a substrate (simulated culture, real culture, or silicon reservoir) with N state units, read by the same feature map ϕ, readout class, regularization search and data budget. Let $\mathcal{E}_N$ be a random

Table 1: Evidential status of each module. *Established*: demonstrated in published work. *Plausible*: follows from established results but not shown for language. *Hypothesis*: proposed here and to be tested.

Module	Status	Basis
Electrode stimulation and spike recording, closed loop	Established	[5, 7]
Culture as fading-memory reservoir	Established	[6, 18]
Predictable vs. unpredictable feedback drives adaptation	Established (games)	[5, 10]
Trained linear readout of spike features	Established	[6, 14]
Token-to-stimulation encoder for language	Plausible	this work
Ring culture as working memory	Hypothesis	this work
Association culture for semantic structure	Hypothesis	this work
Scale-out across racks improves prediction	Hypothesis	this work
Net energy advantage including life support	Hypothesis	this work

echo state network with N units [12].

H0 (null). $\mathcal{E}_N$ performs at least as well as $\mathcal{S}$ on every task. Under **H0**, the biology adds nothing that a cheap random silicon reservoir does not already provide.

H1 (reservoir value). $\mathcal{S}$ beats $\mathcal{E}_N$ on at least one of memory capacity, chaotic-series prediction, or next-character accuracy, across seeds, with a pre-declared margin.

H2 (memory ring). Adding a ring-structured recurrent module increases memory capacity relative to the same neurons wired without the ring.

H3 (closed loop). Critic feedback improves accuracy beyond a frozen substrate whose readout is retrained on the same data, so the gain cannot be attributed to the readout.

H4 (scale-out). Concatenating features from R independent racks improves next-token accuracy more than one rack with R times the neurons.

H5 (energy). Measured joules per predicted token, including perfusion, temperature control and recording hardware, fall below a matched silicon implementation.

Only **H1**–**H4** can be tested in simulation; **H5** requires tissue and instrumentation. A simulation can support or weaken a hypothesis about *designs*. It cannot show that real neurons behave the same way.

5 Simulation-first protocol

Substrate. Leaky integrate-and-fire or Izhikevich neurons [24], 80% excitatory and 20% inhibitory. Cells are placed at random on a two-dimensional "dish" with distance-dependent connection probability and background noise. Pair-based STDP [25] can be switched on or off. The protocol targets 10^3–10^4 neurons in Brian2 [26] or an equivalent simulator.

Encoder and readout. Each symbol drives a fixed random footprint of input neurons for a window of τ ms. Features are spike counts or low-pass spike traces at the end of each window. The readout is ridge regression whose penalty is chosen on a validation split; for classification, the arg-max is taken over one-hot regression targets.

Table 2: Pilot results at equal size (N=500; mean ± s.d. over 5 seeds). Higher is better for MC and accuracy; lower is better for NRMSE. Best substrate per column in bold.

Substrate	Memory capacity ↑	Mackey–Glass NRMSE, h=5 ↓	Next-char accuracy ↑
Input only (no reservoir)	0.10 ± 0.00	0.658 ± 0.002	0.284 ± 0.000
Echo state network (null model)	**49.23 ± 0.62**	**0.005 ± 0.005**	**0.344 ± 0.003**
LIF spiking reservoir (frozen)	0.80 ± 0.07	0.752 ± 0.010	0.316 ± 0.005
LIF, slow synapses (50 ms)	1.44 ± 0.23	0.307 ± 0.013	0.281 ± 0.006
LIF after STDP exposure	1.13 ± 0.09	0.673 ± 0.019	0.307 ± 0.002
Count-based references, same split: unigram / bigram / trigram			0.196 / 0.284 / 0.409

Tasks. (1) Linear memory capacity [15]. (2) Mackey–Glass (τ=17) multi-step prediction [27], scored by normalized RMSE. (3) Next-character prediction on a 27-symbol alphabet, scored by accuracy against unigram, bigram and trigram references.

Baselines and ablations. Required baselines are an equal-size random ESN, an untrained spiking reservoir, an input-only readout (no reservoir), and a small LSTM [28] trained end to end as an upper reference. Required ablations vary neuron count, connectivity, noise, synaptic time constant, plasticity on or off, ring module on or off, and number of racks.

Reporting rules. Hyper-parameters are chosen on a held-out tuning seed with grids of equal size for every substrate. Results are reported as mean ± standard deviation over at least five seeds. Negative results are reported with the same prominence as positive ones. Titles and abstracts state "in silico" wherever no tissue was used.

6 Pilot study

We ran a reduced version of the protocol to check the pipeline and to obtain a first answer to **H0**. All code is pure NumPy, runs on a laptop CPU, and is included with this preprint.

Setup. All reservoirs have N=500 units. *ESN*: $\tanh$ units, 10% connectivity, spectral radius 1.05, input scale 0.1. *LIF*: current-based leaky integrate-and-fire neurons, τ_m=20 ms, τ_{syn}=5 ms, 2 ms refractory period, dt=1 ms, 80/20 E/I with inhibition 4.5× stronger, and Gaussian distance-dependent wiring on the unit square. Each sample is presented for 15 ms. Features are spike traces low-pass filtered with a 20 ms time constant. *LIF-slow* uses τ_{syn}=50 ms. *LIF+STDP* undergoes 1,500 samples of unsupervised exposure with pair-based STDP on E→E synapses and is then frozen. Hyper-parameters were chosen on tuning seed 99 from two 3×3 grids (ESN: spectral radius and input scale; LIF: input gain and noise). Every substrate gets the same ridge readout and the same penalty search. Memory capacity sums k=1 . . . 80. Mackey–Glass prediction uses horizon h=5. Next-character prediction uses the first 40,000 characters of Tiny Shakespeare [29], mapped to 27 symbols (24,000 train, 5,000 validation, about 10,900 test). Results are the mean ± s.d. over 5 seeds.

Results. Table 2 and Figure 2 report the outcome. The equal-size echo state network was the best substrate on all three tasks, so **H0** is not rejected at this scale. On memory capacity the ESN reached

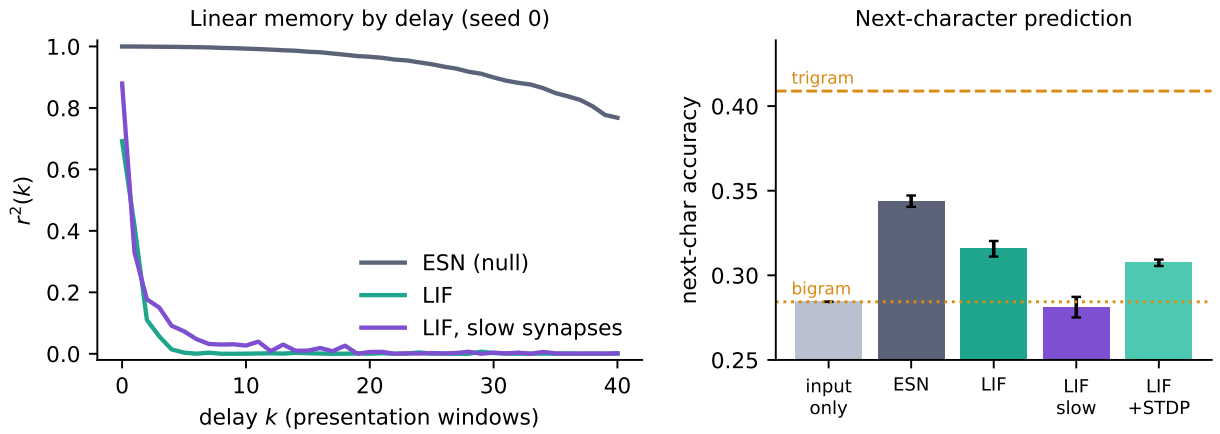


Figure 2: **Left:** delay-resolved linear memory $r^2(k)$ for seed 0. The echo state network keeps information for dozens of steps, while the spiking reservoir loses most of it within three presentation windows (about 45 ms). Slower synapses extend the tail. **Right:** next-character accuracy per substrate (mean $\pm$ s.d.) against count-based references.

49.2, against 0.80 for the frozen LIF reservoir, 1.44 with slow synapses and 1.13 after STDP exposure. On Mackey–Glass the ESN's error (0.005) was roughly sixty times lower than the best spiking variant (0.307, slow synapses). The fast-synapse LIF reservoir (0.752) and its STDP variant (0.673) did worse than a readout of the raw input (0.658), because low-pass spike traces encode the current value only partially. Next-character prediction split the spiking variants. The frozen LIF reservoir (0.316) and its STDP variant (0.307) beat the input-only readout (0.284, which equals the bigram reference), so they carried usable context. Both stayed below the ESN (0.344) and well below a count-based trigram (0.409). Slow synapses, which helped on the two continuous tasks, dropped next-character accuracy to 0.281, just under the input-only readout; their longer integration appears to blur successive symbols. STDP exposure raised memory capacity modestly (0.80 to 1.13) without improving next-character accuracy. Mean firing rates were 5.5 Hz (LIF), 15.1 Hz (after STDP) and 32.2 Hz (slow synapses), so none of the networks was silent or saturated. The pattern points to time scale as the binding constraint: the change that most improved spiking memory also showed that the right time scale depends on the task.

A first look at the memory ring (H2). On one seed we probed a naive ring: excitatory cells were placed in 40 sectors, each projecting to the next. Table 3 shows the result. The weakest ring roughly matched the plain two-dimensional wiring, but only while the network fired at about 0.5 Hz. At $w{=}0.2$ the network fired at a normal rate (7.8 Hz) yet held almost no memory, and stronger coupling pushed firing to 54–148 Hz while still losing it. No ring setting came close to the gain from slower synapses. This is a probe, not a test of **H2**, but it already says something useful: a memory ring will not work by topology alone. It needs a time scale and an excitation–inhibition balance that keep activity both alive and bounded.

7 Discussion

What the pilot does and does not say. The pilot does not show that living tissue is a poor reservoir. Real cortical cultures hold stimulus information for about a second [18], far longer

Table 3: Single-seed probe of a naive memory ring (sum of $r^2(k)$ for $k{=}1\ldots11$, and mean firing rate).

Configuration	Memory ($k{=}1\ldots11$)	Mean rate (Hz)
2D baseline	0.45	5.3
2D, slow synapses (50 ms)	1.10	28.7
ring, w=0.1	0.49	0.5
ring, w=0.2	0.07	7.8
ring, w=0.5	0.14	54.0
ring, w=1.0	0.05	148.0

than our 5 ms-synapse model, because they have mechanisms such as slow receptors, short-term plasticity and adaptation that our point neurons lack. What the pilot does show is sharper. The memory a spiking substrate offers is set by its physical time constants relative to the token rate, whereas a rate-based ESN's memory is set by its spectral radius and is free of wall-clock time. For a wetware language machine, the decisive design variable is therefore the ratio of intrinsic memory time to presentation window, not the number of cells. This can be engineered from both sides: slower presentation (fewer tokens per second) or slower substrates (ring modules, slow synapses, neuromodulated cultures). Both have costs that **H5** must account for.

The input–output bottleneck. A transformer reads its whole context through dense matrix products. A culture is written to through a handful of stimulation sites (eight sensory electrodes in DishBrain, even though the array records from 26,000; eight electrodes per organoid on the Neuroplatform) [7, 10], and it is read through spike counts. In the Cellular Data Center, electrode count and readout dimensionality, not neuron count, bound how much information can enter and leave per token. Scale-out across racks (**H4**) is our proposed answer, and it is testable in simulation before any tissue is used.

Energy, honestly counted. The 20 W figure belongs to a whole brain with its own vasculature. A cultured system pays for perfusion, incubation, amplifiers and the silicon front and back ends. Any energy claim (**H5**) must report joules per token for the complete rack, measured, not inferred from the brain.

Ethics. Work with human-derived neural tissue raises questions about donor consent, sentience-related claims and public communication. We follow the embedded-ethics approach proposed for organoid intelligence [9]. We avoid language that attributes experience to cultures, and we commit to the feedback amplitudes already used in the field.

Limitations. The pilot is small ($N{=}500$, 5 seeds, three tasks). It uses a simplified neuron model and one feature map, and it omits the LSTM reference and the multi-rack ablation that the full protocol requires. Hyper-parameter grids were small and symmetric, and larger searches could change the ranking. Simulation results transfer to tissue only as hypotheses.

8 Conclusion

Living neurons have learned to play games, and they can now be rented. Whether they can carry any part of a language model is an empirical question, and this paper reduces it to a test that any lab can run. The Cellular Data Center gives the question an architecture, five hypotheses give it edges, and a strict silicon null model keeps it honest. Our first pilot did not reject that null. It did show where the effort should go next: matching the clock of the tissue to the clock of language. We release the protocol and code so that the next result, positive or negative, can be checked by others.

Code and data availability

The pilot code (`pilot.py`, `tune.py`, `curves.py`, `ring_probe.py`), raw results in JSON, and the figure script are included as ancillary files with this preprint. The Tiny Shakespeare text is public [29]; the SHA-256 of the file used is `86c4e6aa...565ed`.

Competing interests

The authors declare no competing interests and have no affiliation with Cortical Labs or FinalSpark.

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